

Element C:

Presentation and Justification of Solution Design Requirements

Priority	Design Requirements	Explanation for priority	Meet need of stakeholders	What is measured	Associated Resource
1 (high priority)	Produce sufficient water flow	The main requirement/ necessity of the sink is to be able to dispense water from the storage.	Teachers will be able to use the sink and allow students to interact with expelled water. Students will be able to wash their hands and brush their teeth with the water provided by the sink.	Total Gallons per Minute	How To Troubleshoot Low Water Pressure in Your House.
2 (high priority)	Portability.	The sink needs to be able to be carried in some shape or form by a teacher or teacher aid. This task should be manageable, and not a difficult task.	In our case, a student with a short attention span may need to have a portable sink quickly moved to their desk in order for them to practice.	Weight (lbs) We may also want to measure the time it takes to move the portable sink	How Does Transitioning To A New Location Affect Children with Developmental Disabilities - 24 Hour Home Care
3 (high priority)	Student Functionality	Teachers expressed to us the importance of a student knowing how to function a sink by themselves.	This will meet the needs of the students and the teachers as the main role of this sink is to work as	Success rate of students being able to operate the sink independently.	The Direct Connection Between Learning Experience and

		When asked if the teacher could turn the sink on for the student, the teacher explained that while it would be acceptable, they would much rather prefer a student to be able to independently operate the sink.	a transitional tool to help develop the students ability to use the sink effectively. This will help the teachers in their goal for the students of being able to operate the sink independently.		Retention).
4 (high priority)	Standard Sink appearance	Our sink should, to some extent, resemble a real sink in order to provide an efficient learning tool. This is important because if the product teaches someone how to wash their hands with a device that does not resemble or function like a sink, they may not know how to wash their hands or brush their teeth properly in the real world situation..	Teachers requested that in addition to being a functional everyday sink, it is just as important that this sink teaches the students basic uses of sinks. Our portable sink will need to teach these students how to wash their hands on any sink in the building or at home. Hopefully the sink will facilitate the teachings of staff and learning of students.	Satisfactory level of how easy it is for the teachers to show the students how to use the sink. Rating out of 10 from Stakeholder 1-worst 10- best	The Direct Connection Between Learning Experience and Retention).
5 (high priority)	Safety	The sink should not have any sharp edges to puncture the children, there should also not be any materials that could cause illness such as rusty metal. The sink should be sturdy and heavy	This will meet the needs of the students by preventing injury while they are learning to utilize a transitional sink and being able to transition to a normal sink	The total number of edges or ridges considered sharp or dangerous by stakeholder during preview.	

		enough to not be able to be pushed off and hurt another child. The flow rate and temperature of the sink should be within normal range of other sinks to avoid dangerously hot temperatures and really high flow rates that could damage the skin of a student.	without many differences in safety level. This will meet the needs of the teachers as they will not have to worry as much about the safety of the students and allow them to use the sink by themselves and build the confidence to utilize the regular sink.		
6 (high priority)	Cleanliness	The sink needs to be able to clean easily. This is an important objective due to the illness constantly going around especially in the winter time when hand washing is extremely important to getting rid of germs. The touch on the sink and the overall dirtiness of younger kids creates an area where germs can easily spread person to person. Along with this the teachers have used stronger cleaning materials as of the Covid-19 pandemic to sanitize classroom materials. The sink would have to	This complies with the needs of the children as if the sink is easily cleanable, then the kids will be less likely to pick up germs and get sick. This also follows the needs of the teachers where they have to clean the materials in the classroom to prevent the spread of illness especially in the winter time where illness spreads easier. If the sink can also hold up against the stronger chemicals then teachers will get a more efficient clean.	Amount of visible smears, liquids, and or dirt (Measured in number of spots)	Food Safety Tip: Clean the kitchen sink

		withhold these chemicals without falter in order to work properly.			
7 (low priority)	Storage	Storage in the classroom is a difficult task for the teachers as even though the room for the life skills class is bigger, they also have a lot more stuff to store in the classroom. Along with this the storage of the items must be in a way to not distract the students and allow them to operate on a daily basis without the constant attention being on the sink.	This will meet the needs of the students as if the sink is stored in a way that prevents them from being distracted by the sink all the time then the students will be able to learn more and have a better grasp on the material being taught. This will also meet the needs of the teacher as the sink would be tucked away in the class away from being a distraction and not taking up as much space leading to them being able to teach better to students.	The total volume of the sink	
8 (low priority)	Customizability	This requirement is to help the students learn more while using the sink. During our initial meeting with the teachers we learned that although the main objective is to make a working sink, it would be great if learning elements could be added to help also	This will meet the needs of the students by allowing them to learn new	The total number of pieces that can be adjusted to enhance experience for different students.	The Direct Connection Between Learning Experience and Retention).

		stimulate more learning.			
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